AI-POWERED PERSONALIZED TRAVEL PLANNING SYSTEM

A PROJECT REPORT

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Submitted by

240701074 – BASKAR A

240701420 – RAMCHANDRAN V S

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

RAJALAKSHMI ENGINEERING COLLEGE

THANDALAM, CHENNAI 602 105

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BONAFIDE CERTIFICATE

Certified that this project report "AI-POWERED PERSONALIZED TRAVEL PLANNING SYSTEM" is the bonafide work of "**BASKAR A (240701074) and RAMCHANDRAN V S (240701420)**" who carried out the project work under my supervision.

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ABSTRACT

The AI-Powered Personalized Travel Planning System is designed to simplify the process of planning trips according to individual preferences and budget requirements. Planning a trip involves selecting destinations, transportation, accommodation, tourist attractions, and activities, which can be time-consuming when handled manually. This system uses rule-based Artificial Intelligence to generate personalized travel plans based on the user's destination, budget, travel duration, and interests. It applies predefined rules, logical reasoning, and a knowledge base to recommend suitable places and organize activities into a day-wise itinerary. The system also provides estimated expenses for transportation, accommodation, food, and sightseeing. Unlike traditional AI systems that require Machine Learning or Deep Learning models, this project uses predefined rules and decision-making logic. The proposed system helps travelers save time, organize their trips efficiently, and manage expenses within their available budget.

CHAPTER 1

INTRODUCTION

Travel planning is an important activity that involves selecting destinations, arranging transportation, booking accommodation, and choosing tourist attractions. Travelers often spend a lot of time searching different websites and comparing available options to create a suitable travel plan. Managing all these details manually can be difficult, especially when considering budget, travel duration, and personal interests.

The AI-Powered Personalized Travel Planning System is designed to simplify the travel planning process by providing customized travel itineraries based on individual requirements. The system collects information such as destination, budget, number of travel days, number of travelers, and interests, including nature, adventure, history, shopping, and cultural activities. Based on these details, it recommends suitable tourist attractions and activities.

The system uses rule-based Artificial Intelligence to make decisions using predefined conditions, logical reasoning, and a knowledge base. It selects appropriate attractions, organizes activities into a day-wise schedule, and estimates expenses for transportation, accommodation, food, and sightseeing. Unlike Machine Learning and Deep Learning approaches, this system does not require training data or complex predictive models. Its rules and travel information can be updated whenever necessary.

The main objective of this project is to help travelers plan their trips efficiently, reduce the time required for research, and manage their expenses within the available budget. The system provides a simple and user-friendly interface through which users can generate and review personalized itineraries. Overall, the project demonstrates how rule-based AI techniques can support intelligent decision-making and provide practical solutions for travel planning without using Machine Learning or Deep Learning.

CHAPTER 2

LITERATURE REVIEW

Author & Year	Methodology	Drawback	Improvement in FlowGuard
Kabassi (2010)	Reviews personalization techniques for tourist recommendations based on user preferences.	Travelers may need to provide detailed preference information.	Collects simple inputs such as budget, destination, duration, and interests.
Borràs et al. (2014)	Surveys intelligent tourism recommender systems and their recommendation techniques.	Different recommendation approaches can increase implementation complexity.	Uses a simple rule-based decision engine.
Gavalas et al. (2014)	Reviews mobile tourism recommendation systems and context-aware travel services.	Some systsms depend on mobile technologies and contextual data.	Generates itineraries through a straightforward web interface.
Vansteenwegen and Van Oudheusden (2007)	Studies the Tourist Trip Design Problem and planning visits to multiple attractions.	Planning becomes challenging when time and visit constraints increase.	Uses predefined rules to distribute attractions across travel days.
Huang et al. (2012)	Integrates evidence-based reasoning, Bayesian networks, and mapping for tourist attraction recommendations.	Probabilistic reasoning and integration can require additional implementation effort.	Uses explicit conditions that are easy to understand and maintain.

Author & Year	Methodology	Drawback	Improvement in FlowGuard
Lee et al. (2009)	Uses an ontology-based, multi-agent approach for travel recommendations in Tainan City.	Domain-specific knowledge and ontology maintenance may be complex.	Stores attraction details in a simple JSON/CSV knowledge base.
Lucas et al. (2013)	Presents a hybrid recommendation approach for tourism.	Hybrid approaches may combine multiple techniques and data sources.	Uses one transparent rule-based engine for preference matching.
Gavalas and colleagues (2014)	Examines tourism recommendations for mobile users, including preferences and contextual information.	Mobile and context-aware functionality can add development complexity.	Focuses on essential user inputs and itinerary generation.
Borràs et al. (2014)	Classifies tourism recommendation methods and discusses planning and personalization.	A survey describes existing approaches rather than implementing this project's complete application.	Implements an integrated application for attraction selection, scheduling, and cost estimation.
Proposed System (2026)	Uses rule-based AI, logical conditions, a knowledge base, itinerary generation, and budget calculation.	Results depend on predefined rules and the accuracy of travel information.	Provides personalized day-wise travel plans without Machine Learning or Deep Learning.

Table 2.1 literature survey

Recent research on tourism recommender systems, personalized travel planning, and itinerary-generation techniques was reviewed to understand the current approaches to travel recommendation and identify opportunities for improvement. Table 2.1 summarizes the selected studies, their methodologies, their principal drawbacks, and how the proposed AI-Powered Personalized Travel Planning System addresses those limitations.

A common theme across the reviewed studies is that travel recommendation and itinerary planning can involve complex algorithms, extensive user data, or multiple information sources. Some approaches focus mainly on recommending tourist attractions, while others emphasize route optimization or personalized recommendations without integrating budget management and day-wise itinerary generation into a single simple application. These limitations highlight the need for a travel planning system that combines user preferences, travel duration, attraction selection, and expense estimation in a unified process.

The specific contribution of the proposed AI-Powered Personalized Travel Planning System is to combine personalized attraction selection, rule-based decision-making, day-wise itinerary generation, and budget estimation into a single application. Unlike approaches that depend on Machine Learning or Deep Learning, the proposed system uses predefined rules, logical conditions, and a structured travel knowledge base. This makes the system lightweight, explainable, and easy to maintain. By considering the user's destination, budget, travel duration, and interests, the system aims to generate practical travel plans while helping users save time and manage expenses effectively.

CHAPTER 3

PROPOSED SYSTEM

The AI-Powered Personalized Travel Planning System is designed to generate customized travel itineraries based on the user's destination, budget, travel duration, interests, and preferences. Instead of searching multiple websites and manually organizing travel information, users can enter their travel requirements into the system and receive a suitable day-wise travel plan. The system aims to simplify travel planning, reduce time and effort, and help users manage their expenses effectively.

The system collects information such as the preferred destination, available budget, number of travel days, number of travelers, and interests such as nature, adventure, history, shopping, and cultural activities. It retrieves relevant information from a structured travel knowledge base containing tourist attractions, activity categories, visiting durations, entrance fees, accommodation options, and transportation costs. This information can be stored in JSON or CSV files and updated whenever new destinations or attractions are added.

The rule-based AI engine evaluates the user's preferences using predefined conditions and logical reasoning to recommend suitable attractions. For example, users interested in history are recommended museums and historical monuments, while users with a limited budget are directed toward free or low-cost attractions. The itinerary generation module arranges the selected places into a day-wise schedule based on travel duration and visiting time. The budget calculation module estimates expenses for transportation, accommodation, food, entrance tickets, and other activities. If the estimated cost exceeds the available budget, the system suggests more economical alternatives.

The final travel plan is displayed through a user-friendly interface, showing recommended attractions, daily activities, and estimated expenses. Users can modify their travel preferences and generate an updated itinerary whenever necessary. The system can also provide an option to save travel plans for future reference. Unlike approaches that require Machine Learning or Deep Learning, the proposed system uses predefined rules, logical conditions, and a structured knowledge base. This makes the system simple, explainable, and easy to maintain while helping travelers plan trips efficiently and stay within their estimated budget.

3.1 ARCHITECTURAL DIAGRAM

The architecture of the proposed system can be represented as follows:

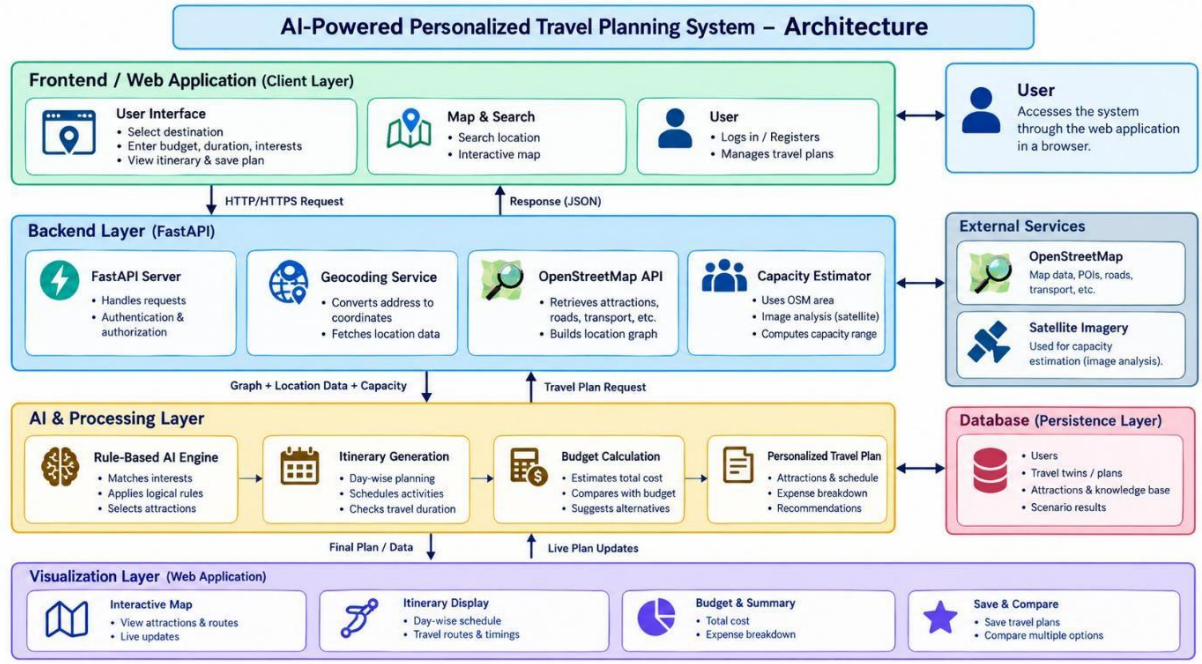


figure 3.1 Architecture diagram

The user enters the destination, budget, travel duration, number of travelers, and interests through the web application. The system validates the input and uses the travel knowledge base and external map services to retrieve information about destinations, attractions, transportation, and accommodation. The Rule-Based AI Engine applies predefined rules to select suitable attractions based on the user's interests and budget. The Itinerary Generation Module organizes the selected attractions and activities into a day-wise travel schedule, while the Budget Calculation Module estimates transportation, accommodation, food, and entry-ticket expenses. If the estimated cost exceeds the user's budget, the system suggests more economical alternatives. Finally, the personalized travel plan, including attractions, routes, schedules, and expense details, is displayed through the web interface. User details, travel preferences, destination information, and saved itineraries are stored in the database for future access

CHAPTER 4

MODULES DESCRIPTION

The proposed system consists of six modules. Each module performs a specific task and works with the other modules to generate a personalized travel plan.

4.1 MODULE 1 – USER INPUT AND PREFERENCE COLLECTION

This module collects the user's travel requirements through a simple interface. The user enters the destination, available budget, number of days, number of travelers, preferred activities, and accommodation preferences.

The collected information is passed to the next module for validation and processing. This module ensures that the system understands the user's basic travel needs before generating recommendations.

4.2 MODULE 2 – INPUT VALIDATION

This module checks whether the information entered by the user is valid and complete. It verifies that the budget is a positive amount, the travel duration is greater than zero, and the required destination and preference fields are provided.

If any information is missing or invalid, the system displays an appropriate message and asks the user to correct the input. This prevents errors during itinerary generation.

4.3 MODULE 3 – TRAVEL KNOWLEDGE BASE

The knowledge base stores information about destinations, tourist attractions, activity categories, visiting durations, estimated entrance fees, and other travel expenses.

The information can be maintained using JSON or CSV files. Each attraction can contain details such as its name, location, category, estimated cost, and recommended visiting time.

For example, a destination may contain beaches, historical monuments, parks, museums, and shopping areas. The system retrieves suitable information from the knowledge base according to the user's interests.

4.4 MODULE 4 – RULE-BASED AI DECISION ENGINE

This is the main AI module of the project. It uses predefined rules and logical conditions to select attractions that match the user's interests and budget.

For example:

- IF the user's interest is nature, THEN select parks, gardens, and scenic locations.
- IF the user's interest is history, THEN select museums and historical monuments.
- IF the budget is low, THEN prioritize free or low-cost attractions.
- IF the trip duration is two days, THEN select attractions that can reasonably fit into a two-day itinerary.

The module evaluates these conditions and creates a list of suitable attractions. It does not train a model or use Machine Learning or Deep Learning.

4.5 MODULE 5 – ITINERARY GENERATION AND BUDGET CALCULATION

The module also estimates the cost of transportation, accommodation, food, entrance tickets, and other activities. It calculates the approximate total trip expense and compares it with the user's budget.

If the cost exceeds the available budget, the system can apply rules to replace expensive activities with lower-cost alternatives.

4.6 MODULE 6 – TRAVEL PLAN DISPLAY

This module presents the final travel plan in a clear and user-friendly format. It displays the destination, daily activities, recommended attractions, estimated costs, and total budget.

Users can review the generated itinerary and adjust their interests, budget, or travel duration to obtain an updated plan. The module helps users understand the complete trip schedule before traveling.

CHAPTER 5

CHAPTER 5: IMPLEMENTATION AND RESULTS

5.1 Implementation

The AI-Powered Personalized Travel Planning System is developed using Python and Flask for backend processing, with HTML, CSS, and JavaScript for the user interface. A JSON/CSV knowledge base stores tourist attractions, destination details, estimated costs, and activity information. The system uses rule-based AI techniques to recommend destinations and generate travel itineraries without machine learning or deep learning.

5.2 System Modules

The system consists of several modules, including user input, input validation, knowledge base management, rule-based recommendation, itinerary generation, budget calculation, and result display. Each module performs a specific task to ensure the travel plan is generated according to the user's requirements.

5.3 Rule-Based Recommendation

The rule-based engine analyses the user's destination, travel duration, budget, interests, and preferences. Predefined conditions are applied to select suitable tourist attractions. For example, users interested in nature are recommended parks, hills, and waterfalls, while history enthusiasts may receive recommendations for museums and historical monuments.

5.4 Itinerary and Budget Generation

After selecting suitable attractions, the system arranges them into a day-wise itinerary based on the trip duration. It estimates expenses for transportation, accommodation, food, and entry tickets. If the estimated cost exceeds the user's budget, the system can suggest economical attractions or lower-cost travel options..

5.5 Expected Outcomes

The proposed system aims to reduce manual travel-planning effort, simplify destination selection, and improve budget management. It provides an organized itinerary, estimated travel expenses, and recommendations based on user preferences. The system can also be expanded by adding more destinations, attractions, and updated cost information.

5.2 RESULTS

Figure 5.2.1:Personalized Travel Planner interface for entering travel dates, origin, group size, and traveler type to generate customized itineraries.

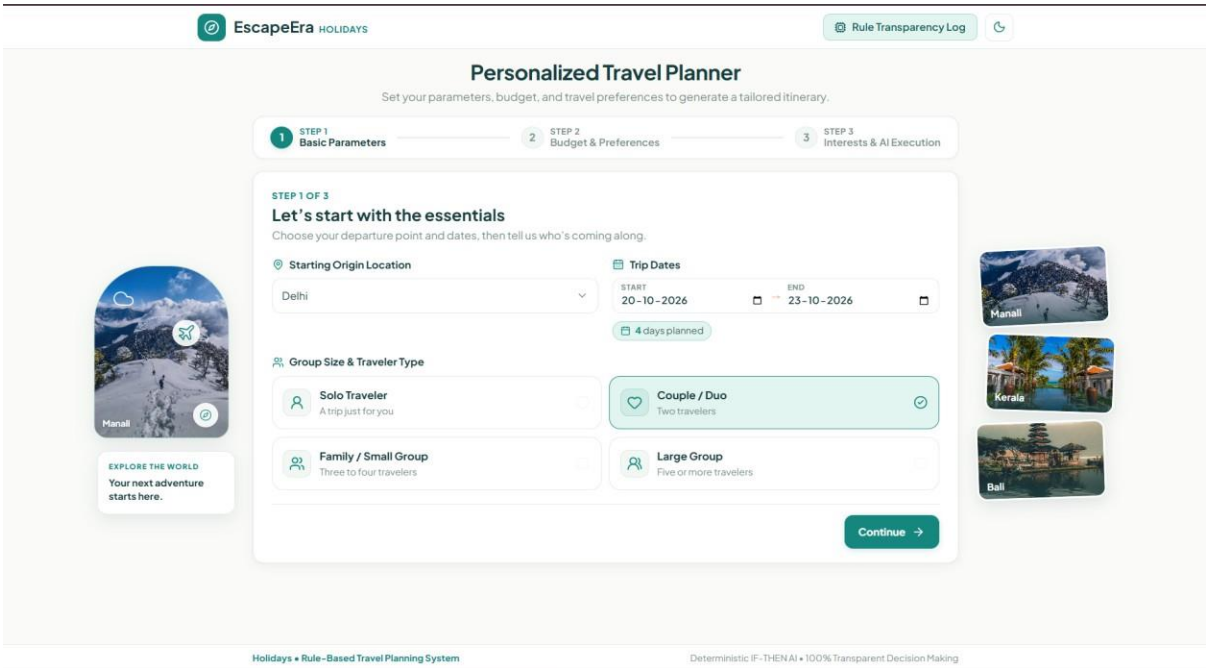


Figure 5.2.1 Basic Parameters Dashboard

Fig. 5.2.2: The interface allows users to set their total budget, preferred transportation mode, and accommodation type for personalized travel planning.

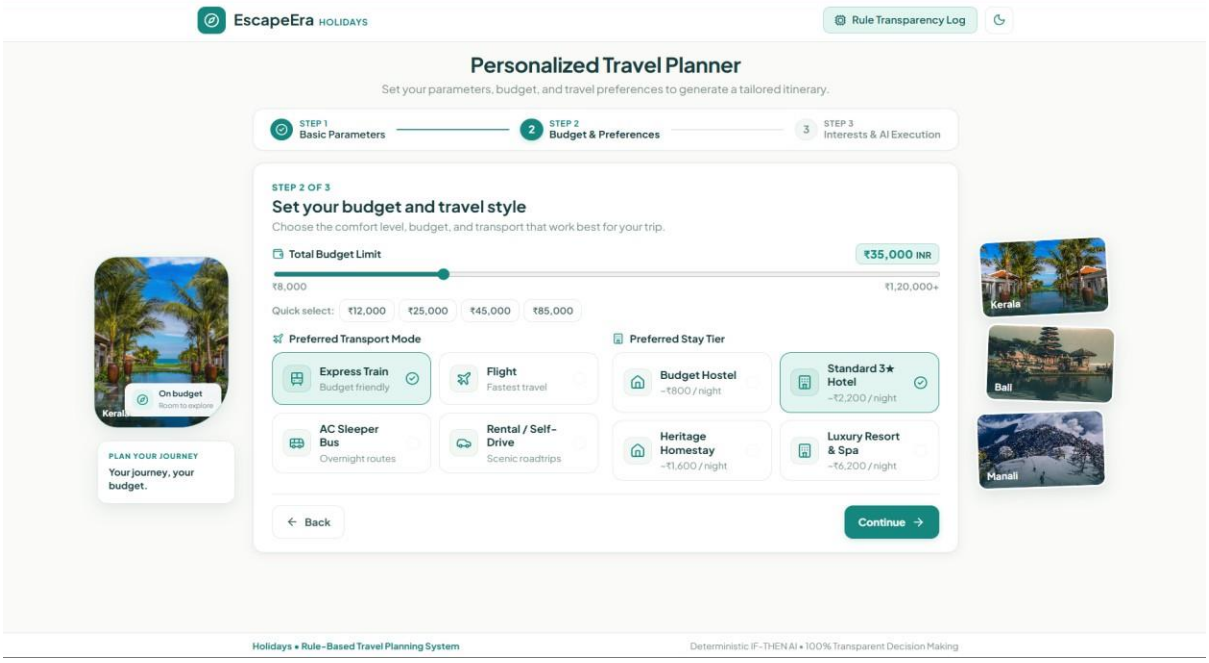


Figure 5.2.2 Budget Preferences Dashboard

Figure 5.2.3: The dashboard allows users to select travel interests and review trip details before generating a personalized itinerary based on predefined rules.

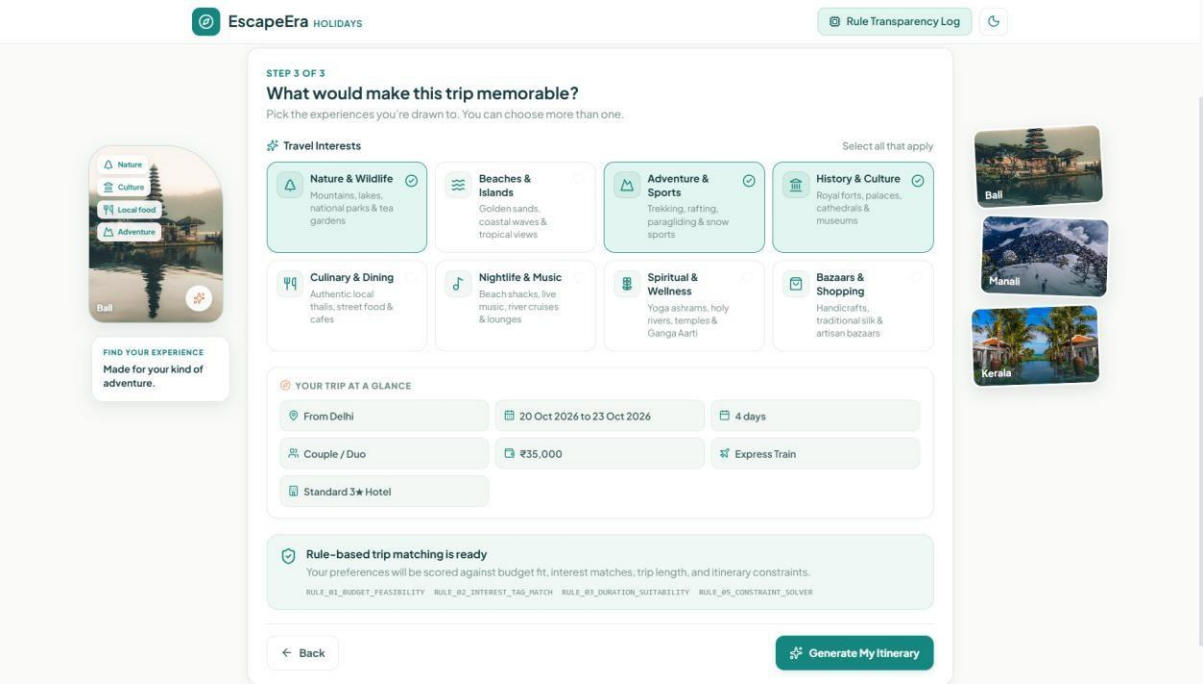


Figure 5.2.3 Travel Interests Selection Dashboard

Figure 5.2.4: The dashboard displays destinations ranked by rule-based matching scores, considering user interests, budget and estimated daily costs.

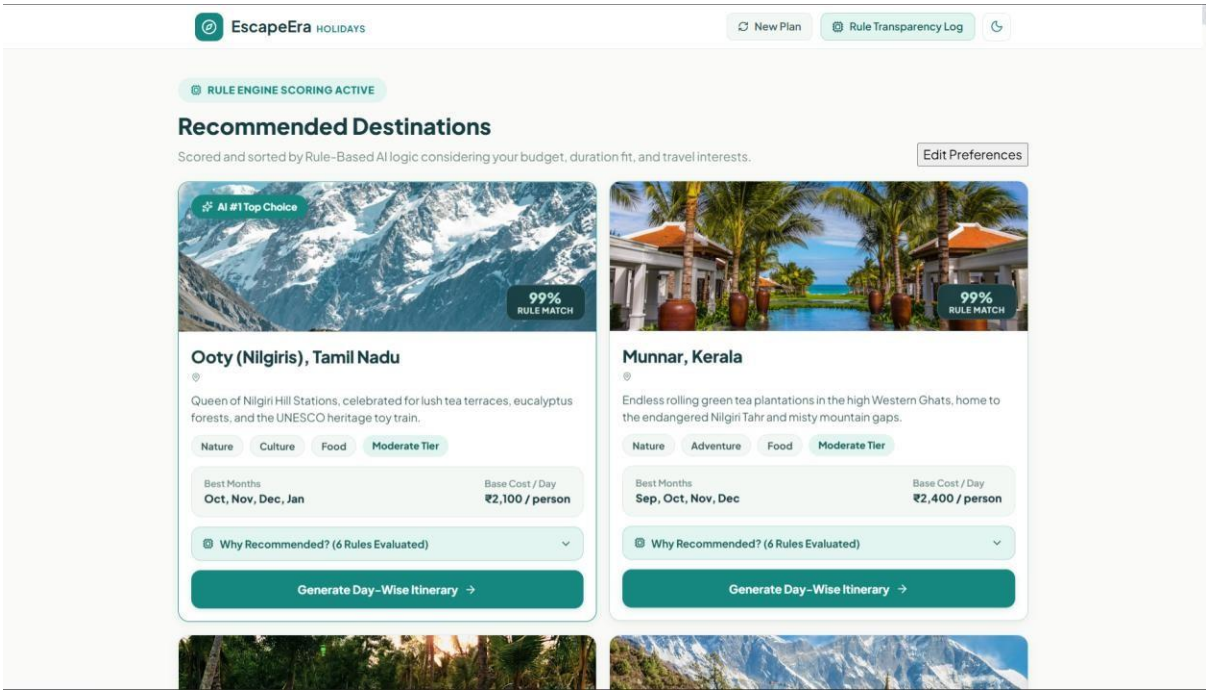


Figure 5.4.1: Rule-Based Destination Recommendation Dashboard

The application collects information such as the starting location, travel dates, number of travellers, budget, transportation mode, and accommodation type. Users can select multiple travel interests, including nature and wildlife, adventure and sports, history and culture, beaches, food, and shopping. These preferences are processed by the Rule-Based AI Engine to identify destinations that match the user's requirements.

The destination recommendation dashboard displays suitable destinations with descriptions, interest categories, recommended travel seasons, estimated daily costs, and rule-matching scores. Users can review the recommended destinations and view the reasons behind each recommendation through the rule transparency feature. This improves the explainability of the system by showing how predefined rules evaluate budget suitability, interest matching, and travel duration.

After selecting a destination, users can generate a personalized day-wise itinerary. The system organizes travel activities according to the selected dates, preferences, and estimated expenses. It helps users understand the approximate cost of their trip and compare destination options before making travel decisions. The interface also allows users to modify their preferences or start a new travel plan when required.

The proposed system uses rule-based Artificial Intelligence rather than Machine Learning or Deep Learning. It applies predefined conditions and logical rules to evaluate destinations and generate recommendations. Overall, EscapeEra Holidays aims to simplify travel planning by combining preference selection, destination recommendations, transparent rule-based matching, and itinerary generation in one application. The system helps users save time, explore suitable destinations, and plan trips according to their interests and budget.

CHAPTER 6

CONCLUSION AND FUTURE WORK

The AI-Powered Personalized Travel Planning System provides an efficient and user-friendly solution for planning trips according to individual preferences. The system considers factors such as destination, budget, travel duration, number of travelers, and personal interests to generate a suitable day-wise travel itinerary. It reduces the time and effort required for manual travel planning by organizing tourist attractions, activities, routes, and estimated expenses in a single plan.

The proposed system uses a Rule-Based AI Engine that applies predefined rules to recommend suitable attractions based on users' interests and budgets. The budget calculation module estimates expenses for transportation, accommodation, food, and entry tickets. If the estimated cost exceeds the available budget, the system can suggest more economical alternatives. Since the system uses predefined rules and a structured knowledge base rather than machine learning or deep learning, it is simple to understand, maintain, and modify. Overall, the system helps users make better travel decisions and manage their travel expenses effectively.

In the future, the system can be enhanced by integrating real-time weather updates, transportation information, hotel availability, and current ticket prices. GPS navigation and interactive maps can provide better route guidance. Additional features such as group travel planning, multilingual support, online booking links, downloadable itineraries, and emergency contact information can also be introduced. The knowledge base can be expanded to include more destinations, tourist attractions, and accommodation options. A mobile application could further improve accessibility and convenience. These enhancements can make the system more useful, flexible, and reliable for travelers.

CHAPTER 7

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